Differentiation Rules: Power, Product, Quotient, Chain

ANSWER KEY

Power, sum, and constant rules

1 Differentiate each function:

a $f(x) = 5x^4 - 3x^2 + 7$
$= 20x^3 - 6x$

b $f(x) = \frac{1}{x^3} \quad = -3x^{-4}$

c $f(x) = 4\sqrt{x} \quad = \frac{2}{\sqrt{x}}$

Product and quotient rules

2 Differentiate $f(x) = (2x+1)(x^2-3)$ using the product rule.

$f'(x) = 2(x^2-3) + (2x+1)(2x) = 2x^2 - 6 + 4x^2 + 2x = 6x^2 + 2x - 6$.

3 Differentiate $f(x) = \frac{x^2+1}{x-2}$ using the quotient rule.

$f'(x) = \frac{2x(x-2) - (x^2+1)(1)}{(x-2)^2} = \frac{x^2-4x-1}{(x-2)^2}$.

4 Differentiate $f(x) = x^2 \sin x$.

$f'(x) = 2x\sin x + x^2\cos x$.

The chain rule

5 Differentiate each function using the chain rule:

a $f(x) = (3x^2-5)^4$
$= 4(3x^2-5)^3(6x) = 24x(3x^2-5)^3$

b $f(x) = \sin(2x^3) \quad = 6x^2\cos(2x^3)$

c $f(x) = e^{-x^2} \quad = -2xe^{-x^2}$

d $f(x) = \ln(x^2+1) \quad = \frac{2x}{x^2+1}$

6 Differentiate $f(x) = \sqrt{4x + \cos x}$.

$f'(x) = \frac{4 - \sin x}{2\sqrt{4x+\cos x}}$.